

Forces and motion

It can be difficult to make an object move, but once it is moving, it will go on moving until something stops it. Force is needed to start something moving, make it move faster, and make it stop.

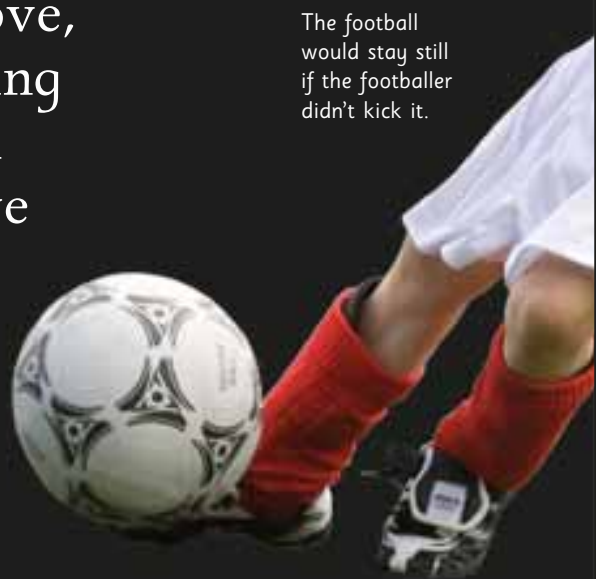
Newton's laws of motion

In 1687, Sir Isaac Newton worked out three important rules that explain how forces make things move. They have become the foundation of physics and work for just about everything, from footballs to frogs.



Newton's second law

The bigger the force and the lighter the object, the greater the acceleration. A professional cyclist with a lightweight bike will accelerate faster than a normal person cycling to work.



The football would stay still if the footballer didn't kick it.

Newton's first law

An object stays still if it isn't being pushed or pulled by a force, or it keeps moving in a straight line at a constant speed.



Forces make things accelerate. The force is created by the cyclist's powerful legs.

Newton's third law

Every action has an equal and opposite reaction. The leaf moves away as the frog leaps in the opposite direction.